Day 4: Other Protocols in Action

FTP Server Setup, SNMP Use Cases, and FTP vs HTTP

Objective

The objective of this assignment is to understand FTP and SNMP, set up an FTP server, transfer a file using FileZilla, and compare FTP with HTTP.

Theory: FTP and SNMP

1. FTP (File Transfer Protocol)

FTP is a network protocol used to transfer files between a client and a server. A user can upload files to a server or download files from it. Standard FTP commonly uses TCP port 21 for control connections; active and passive data connections use additional ports.

2. SNMP (Simple Network Management Protocol)

SNMP is used by network administrators to monitor and manage network devices such as routers, switches, servers, printers, and access points. It can report device status, interface traffic, errors, CPU or memory usage, and alerts. Traditional SNMP commonly uses UDP port 161 for queries and UDP port 162 for traps/notifications.

Task 1: Install and Configure FileZilla Server

1. Download FileZilla Server from the official FileZilla website and install it on your Windows computer.
2. Open FileZilla Server and connect to its administration interface.
3. Create a user account in the Users section and set a password. Use a strong password and do not share it publicly.
4. Add a shared directory (shared folder) and grant the new user the required read/write permissions for the practical.
5. Check the server's listening port and Windows Firewall rules. FTP commonly uses port 21 for control; data connections may require passive-mode port settings.
6. For a same-computer test, connect to 127.0.0.1. For another device on the same LAN, use the server computer's local IP address. Do not expose an FTP server directly to the public Internet for this classroom exercise.

Task 2: Connect with FileZilla Client and Upload an Image

7. Install FileZilla Client on the computer used for the test.
8. Open Site Manager and create a new site.
9. Enter the server host (127.0.0.1 for the same computer, or the server's LAN IP for another device), port 21, and the FTP username and password created in Task 1.
10. Choose the protocol and encryption setting supported by your server. Plain FTP does not encrypt usernames, passwords, or files; use FTPS where available and keep the test on a trusted local network.
11. Connect. In the left panel, find an image file on your computer. In the right panel, open the server's shared folder.
12. Drag the image from the left panel to the right panel to upload it. Confirm that the transfer finishes successfully.

Screenshot placeholder: FileZilla Client showing a successful image upload.

Task 3: Download and Verify the File

13. In FileZilla Client, locate the uploaded image in the server's shared folder (right panel).
14. Choose a different local folder in the left panel, such as a new folder named FTP_Download_Test.
15. Download the image from the server to that folder.
16. Compare the original and downloaded file sizes in File Explorer (right-click the file and choose Properties). For a stronger check, compare the SHA-256 hashes using PowerShell: Get-FileHash "C:\path\to\file.jpg" -Algorithm SHA256. Run the command for both files and compare the Hash values.

Screenshot placeholder: Downloaded image in the different folder and its Properties window.

Conclusion: The transfer is successful if the downloaded file opens correctly and its size matches the original. Matching SHA-256 hashes provide stronger verification.

Task 4: Three Real-World Uses of SNMP

Scenario	How SNMP is used	Benefit
1. ISP network monitoring	An Internet Service Provider can monitor routers, switches, and link traffic to identify high bandwidth use or interface errors.	Helps detect congestion and network faults.
2. Campus Wi-Fi management	A campus IT team can monitor wireless access points and switches for uptime, traffic, and device availability.	Helps locate failed access points and plan capacity.
3. Website or data-centre operations	IT staff can monitor server and network-device health, interface traffic, and hardware alerts in a data centre that hosts websites or apps.	Helps improve availability and troubleshoot issues.

Task 5: FTP vs HTTP Comparison

Point	FTP	HTTP
Full form	File Transfer Protocol	Hypertext Transfer Protocol
Common port	TCP 21 for control; data uses separate connections/ports	TCP 80 for HTTP; HTTPS commonly uses TCP 443
Main purpose	Uploading and downloading files	Delivering web pages, APIs, images, and other web resources
Transfer model	Separate control connection and data connection	Request-response messages between client and server
Security	Plain FTP is not encrypted; FTPS adds TLS. SFTP is a different protocol over SSH.	HTTP is not encrypted; HTTPS protects traffic using TLS
Authentication	May use a username and password, or anonymous access if configured	Can use cookies, tokens, Basic/Digest or other application-level methods; public pages may need no login
Typical use cases	Moving files to/from a server, legacy file exchange	Web browsing, web applications, and API communication
Browser support	Not the normal method for modern web browsing	Core protocol used by web browsers and web services

Conclusion

FTP is designed for file transfer, while HTTP is designed for communication between web clients and servers. SNMP helps administrators monitor network devices and identify problems. In this practical, I configured an FTP user and shared folder, uploaded and downloaded an image, and checked the file size. I also studied three uses of SNMP and compared FTP with HTTP.

Practical Evidence Checklist

☐ FileZilla Server installed and user/shared folder created

☐ FileZilla Client connected and image uploaded

☐ Image downloaded to a different folder

☐ Original and downloaded file sizes recorded

☐ Screenshots added from my own practical work

Note: Replace blank fields with your actual results and add your own screenshots. Do not claim that a practical step was completed until you have performed it. FileZilla menu names can vary slightly by version.